MATH 231 – Linear Algebra I

Homework 3 ▪ Matrices and Matrix Multiplication

Kiely Hall 326 ▪ Instructor: Kevin Bedoya

Due: Sunday, November 8, 2026, 11:59 PM.

Lecture	Sections	Topics
Matrix Unit, Part 1	§2.3, §2.5, §3–§9	matrices, entries and dimensions; shapes and patterns; adding and scaling; the transpose and symmetric matrices; inner and outer products; matrix multiplication; matrix–vector products by rows and by columns; the trace

Grading. Every problem is worth the same number of points.

Section numbers. A bare section number such as §8.4 refers to the matrix unit, *Linear Systems and Matrix Notation*, whose numbering restarts at 1. References into the vector unit say so.

On the proof problems (A.4(b), B.1(e), B.2(d), B.3(a), (b), (d), C.2(a), (c), (d), D.3(c), D.4(d), and D.5(c)). None of them requires a clever idea. What they require is that you write down what you are assuming, work entry by entry or from the stated laws, justify each equality by naming the fact that licenses it, and end with a sentence saying what you have shown. A chain of equalities with no words is not a proof; neither is a paragraph of words with no computation. To show two matrices are equal, show that they have the same shape and that their (i, j) entries agree for every i and j.

Collaboration is encouraged — work at a board together, argue about the problems. What you submit must be written by you, in your own words.

What to submit

Push your written work for Parts A–D to your own MATH 231 GitHub repository, as a single PDF named `hw3.pdf` in a folder named `homework/hw3/`. Handwritten and scanned or photographed is completely fine, as long as it is legible and the pages are in order.

PDF is the only accepted format. A `.docx`, a `.jpg`, a `.png`, a `.heic`, or a link to a file hosted somewhere else will not be graded. Phone scanning apps export PDF directly, and every word processor can "Save as PDF".

Type Parts A–D up instead and you earn +5 points of extra credit on this assignment. Any typesetting system counts — LaTeX, Word's equation editor, or anything else that produces readable mathematics — and the extra credit is for the whole of Parts A–D, not for typing up some of it. The file you submit must still be a PDF. Sketches may be drawn by hand and scanned into the PDF.

If you have not set up the repository yet, Setup Guide 1 (§ "Create your folders" and § "Save your work and send it to GitHub") is the walkthrough. Only pushed work can be graded.

Label every answer

Label each answer with the number of the problem it answers — `A.3(d)`, `C.2(b)`, `D.2(e)` — and put your answers **in the same order as they appear in this assignment**. A grader who cannot tell which question a piece of work belongs to cannot award it credit, and that is the most common way a correct answer loses points.

You do not have to copy the question out. This applies to handwritten and typed submissions alike: the label alone is enough. Write `A.3(d)`, then your work. Do not reproduce the wording of the problem.

Part A. Matrices, entries, and patterns

Matrix Unit, §3–§4. An $m \times n$ matrix has m rows and n columns, and its entry in row i and column j is a_{ij} — row first, column second (§3.1). $\mathbb{R}^{m\times n}$ is the set of all $m \times n$ matrices with real entries (§3.2). A $1 \times n$ matrix is a **row vector** and an $m \times 1$ matrix a **column vector** (§3.4). The **main diagonal** of a square matrix is $a_{11}, a_{22}, \ldots, a_{nn}$, and the patterns of §4 — square, rectangular, diagonal, identity, upper and lower triangular, banded and tridiagonal, sparse and dense — are summarized in the table of §4.7.

Problem A.1 (Reading a matrix)**.**
Let

$$\mathbf{A} = \begin{bmatrix} 2 & -1 & 0 & 5 \\ 3 & 4 & -2 & 1 \\ 0 & 7 & 6 & -3 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 1 & 0 \\ -4 & 9 \\ 2 & 2 \\ 8 & -5 \end{bmatrix}, \qquad \mathbf{c} = \begin{bmatrix} 3 \\ -1 \\ 4 \end{bmatrix}, \qquad \mathbf{d} = \begin{bmatrix} 6 & 0 & -2 & 1 & 5 \end{bmatrix}.$$

(a) Give the dimensions of each, the set $\mathbb{R}^{m\times n}$ it belongs to, and its number of entries.

(b) Read off a_{23}, a_{32}, a_{14}, b_{41}, b_{22}, c_{31} and d_{15}.

(c) Which of the addresses a_{41}, b_{23}, c_{13} and d_{51} exist? For each one that does not, say why not.

(d) Write row 2 of $\mathbf{A}$ as a row vector and column 3 of $\mathbf{A}$ as a column vector.

(e) Read $\mathbf{A}$ by rows and by columns, as in §3.5: how many row vectors does it stack, and in which $\mathbb{R}^k$ does each one lie? How many column vectors, and in which $\mathbb{R}^k$?

Problem A.2 (Building a matrix from a rule)**.**
Write out each matrix entry by entry.

(a) $\mathbf{A} \in \mathbb{R}^{3\times 4}$ with $a_{ij} = i - j$.

(b) $\mathbf{B} \in \mathbb{R}^{4\times 4}$ with $b_{ij} = \min(i, j)$.

(c) $\mathbf{C} \in \mathbb{R}^{4\times 4}$ with $c_{ij} = (-1)^{i+j}$.

(d) $\mathbf{D} \in \mathbb{R}^{4\times 4}$ with $d_{ij} = 1$ if $|i - j| = 1$, and $d_{ij} = 0$ otherwise.

(e) $\mathbf{E} \in \mathbb{R}^{4\times 4}$ with $e_{ij} = ij$ if $i \le j$, and $e_{ij} = 0$ if $i > j$.

(f) $\mathbf{F} \in \mathbb{R}^{2\times 5}$ with $f_{ij} = 10i + j$. What do you notice about each entry and its address?

(g) For each of **B**, **C**, **D** and **E**, list every pattern from the table of §4.7 that it has, leaving out sparse and dense, and count its nonzero entries.

(h) Give a rule for a_{ij} that produces the identity $\mathbf{I}_n$, and one that produces the $n \times n$ upper triangular matrix whose entries on and above the diagonal are all 1.

Problem A.3 (Naming the pattern)**.**
For each matrix, list every word that applies to it from: *square, rectangular, diagonal, identity, upper triangular, lower triangular, tridiagonal.* Recall that §4 defines diagonal, identity and triangular matrices only among square matrices, and that a matrix is tridiagonal when $a_{ij} = 0$ whenever $|i - j| > 1$.

(a) $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ (d) $\begin{bmatrix} 4 & 0 & 0 \\ 1 & 0 & 0 \\ 2 & 3 & 0 \end{bmatrix}$

(e) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ (f) $\begin{bmatrix} 1 & 2 & 0 & 0 \\ 3 & 4 & 5 & 0 \\ 0 & 6 & 7 & 8 \\ 0 & 0 & 9 & 1 \end{bmatrix}$ (g) $\begin{bmatrix} 5 \end{bmatrix}$

(h) True or false? Justify each answer from the definitions in §4, and give a counterexample for each false statement.

(i) Every diagonal matrix is upper triangular.

(ii) Every identity matrix is tridiagonal.

(iii) An upper triangular matrix whose diagonal entries are all 0 is the zero matrix.

(iv) If **A** is lower triangular, then so is $c\mathbf{A}$ for every scalar c.

Problem A.4 (The diagonal of any matrix, and counting entries)**.**
The main diagonal makes sense for a rectangular matrix too: for $\mathbf{A} \in \mathbb{R}^{m\times n}$ it consists of the entries $a_{11}, a_{22}, a_{33}, \ldots$, continuing for as long as both row i and column i exist.

(a) List the main diagonal of each matrix:

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 6 & 7 & 8 & 9 & 10 \end{bmatrix}, \qquad \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ 10 & 11 & 12 \end{bmatrix}, \qquad \begin{bmatrix} 3 & 1 & 4 & 1 & 5 & 9 \end{bmatrix}, \qquad \begin{bmatrix} 2 \\ 7 \\ 1 \\ 8 \end{bmatrix}.$$

(b) Show that an $m \times n$ matrix has exactly $\min(m, n)$ entries on its main diagonal.

(c) In an $n \times n$ matrix, count the entries that lie on the main diagonal, strictly above it, strictly below it, and on or above it. Give each count as a formula in n, and check your formulas against a drawing of a 4×4 grid.

Hint. The entries strictly above the diagonal in row i are those in columns $i+1, \dots, n$. Add up over the rows, and use the sum formula for $1 + 2 + \cdots + k$.

(d) The *largest* number of entries that can be nonzero in an $n \times n$ matrix that is diagonal is n. Find the corresponding largest number for an upper triangular matrix and for a tridiagonal matrix.

(e) For a tridiagonal $n \times n$ matrix, compute the fraction of its n^2 entries that can be nonzero, for $n = 4$, $n = 10$ and $n = 1000$. What happens to this fraction as $n \to \infty$?

(f) A 1000×1000 matrix has 4000 nonzero entries. What fraction of its entries are zero? Storing the full grid takes one number per entry. Storing only the nonzero entries takes three numbers per nonzero entry: its row, its column, and its value. How many numbers does each way take? By the working test of §4.6, is this matrix sparse?

Part B. Adding, scaling, and the transpose

Matrix Unit, §5–§6. Matrices of the same shape are added and subtracted entry by entry, $(\mathbf{A} \pm \mathbf{B})_{ij} = a_{ij} \pm b_{ij}$, and scaled entry by entry, $(c\mathbf{A})_{ij} = ca_{ij}$. The laws of addition and scaling are **A1**–**A4** and **S1**–**S4**, introduced in §5.4 ("The algebraic laws"). The transpose of $\mathbf{A} \in \mathbb{R}^{m \times n}$ is $\mathbf{A}^{\mathsf{T}} \in \mathbb{R}^{n \times m}$ with $(\mathbf{A}^{\mathsf{T}})_{ij} = a_{ji}$ (§6.1), and its laws are

$$\textbf{T1}\ \ (\mathbf{A}^{\mathsf{T}})^{\mathsf{T}} = \mathbf{A}, \qquad \textbf{T2}\ \ (\mathbf{A} + \mathbf{B})^{\mathsf{T}} = \mathbf{A}^{\mathsf{T}} + \mathbf{B}^{\mathsf{T}}, \qquad \textbf{T3}\ \ (c\mathbf{A})^{\mathsf{T}} = c\,\mathbf{A}^{\mathsf{T}}$$

(§6.4). A matrix is **symmetric** if $\mathbf{A}^{\mathsf{T}} = \mathbf{A}$ (§6.5).

Problem B.1 (Matrix arithmetic)**.**
Let

$$\mathbf{A} = \begin{bmatrix} 1 & -2 & 0 \\ 3 & 4 & -1 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 2 & 0 & 5 \\ -1 & 3 & 2 \end{bmatrix}, \qquad \mathbf{C} = \begin{bmatrix} 1 & 4 \\ -2 & 0 \\ 3 & 1 \end{bmatrix}.$$

(a) Compute $\mathbf{A} + \mathbf{B}$, $\mathbf{A} - \mathbf{B}$, $-\mathbf{B}$ and $3\mathbf{A} - 2\mathbf{B}$.

(b) Which of $\mathbf{A} + \mathbf{C}$, $\mathbf{A} + \mathbf{C}^{\mathsf{T}}$, $\mathbf{A}^{\mathsf{T}} + \mathbf{C}$ and $\mathbf{B} - \mathbf{C}^{\mathsf{T}}$ are defined? Compute each one that is, and for each one that is not, say why.

(c) Compute $\mathbf{A}^{\mathsf{T}}$, $\mathbf{B}^{\mathsf{T}}$, $(\mathbf{A} + \mathbf{B})^{\mathsf{T}}$ and $\mathbf{A}^{\mathsf{T}} + \mathbf{B}^{\mathsf{T}}$, and confirm **T2** for this pair.

(d) Find the matrix $\mathbf{X}$ with $3\mathbf{X} - \mathbf{B} = \mathbf{X} + \mathbf{A}$. Solve for $\mathbf{X}$ first, naming the law from §5.4 used at each step, and only then substitute.

(e) Find scalars α and β with

$$\alpha \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \beta \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ -2 & 3 \end{bmatrix}.$$

Then show that $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ cannot be written in this form for any α and β.

Problem B.2 (Transposes and symmetric matrices)**.**

(a) Write down the transpose of each matrix, with its shape:

$$\begin{bmatrix}1 & 2 & 3 & 4\end{bmatrix}, \qquad \begin{bmatrix}5\\-1\\2\end{bmatrix}, \qquad \begin{bmatrix}0 & 1\\2 & 3\\4 & 5\end{bmatrix}, \qquad \begin{bmatrix}1 & 2 & 3\\2 & 5 & 4\\3 & 4 & 6\end{bmatrix}.$$

Which of the four is symmetric?

(b) Find every x, y and z for which $\begin{bmatrix}1 & x & 3\\4 & 2 & y\\z & -5 & 0\end{bmatrix}$ is symmetric.

(c) Fill in the entries marked $*$ so that the matrix is symmetric: $\begin{bmatrix}2 & * & *\\7 & -1 & *\\0 & 3 & 4\end{bmatrix}$.

(d) Prove that if $\mathbf{A}$ and $\mathbf{B}$ are symmetric $n \times n$ matrices and c is a scalar, then $\mathbf{A} + \mathbf{B}$ and $c\mathbf{A}$ are symmetric. Name the law from §6.4 used at each step.

Problem B.3 (Splitting a square matrix into symmetric and skew-symmetric parts)**.**
A square matrix $\mathbf{K}$ is **skew-symmetric** if $\mathbf{K}^\mathsf{T} = -\mathbf{K}$, that is, if $k_{ji} = -k_{ij}$ for every i and j. In (a), (b) and (d), work from the definitions and the laws **T1**–**T3**, **A1**–**A4** and **S1**–**S4**, naming each one you use.

(a) Prove that every diagonal entry of a skew-symmetric matrix is 0.

(b) Prove that for every square matrix $\mathbf{A}$, the matrix $\mathbf{A} + \mathbf{A}^\mathsf{T}$ is symmetric and the matrix $\mathbf{A} - \mathbf{A}^\mathsf{T}$ is skew-symmetric.

(c) It follows that every square matrix is the sum of a symmetric matrix and a skew-symmetric one:

$$\mathbf{A} = \underbrace{\tfrac{1}{2}\big(\mathbf{A} + \mathbf{A}^\mathsf{T}\big)}_{\text{symmetric}} + \underbrace{\tfrac{1}{2}\big(\mathbf{A} - \mathbf{A}^\mathsf{T}\big)}_{\text{skew-symmetric}}.$$

Compute both parts for $\mathbf{A} = \begin{bmatrix}1 & 4 & -2\\0 & 3 & 5\\6 & -1 & 2\end{bmatrix}$, and check that they add up to $\mathbf{A}$.

(d) Prove that the only matrix that is both symmetric and skew-symmetric is the zero matrix.

Part C. Inner and outer products

Matrix Unit, §7. From §6.3 on, a vector $\mathbf{v} \in \mathbb{R}^n$ is a column, an $n \times 1$ matrix, and $\mathbf{v}^\mathsf{T}$ is the same vector written as a row. The bracket notation $[\,v_1, \ldots, v_n\,]$ is still used inline, and means the column. For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ the **inner product** $\mathbf{u}^\mathsf{T}\mathbf{v} = \sum_{k=1}^{n} u_k v_k$ is the dot product, a 1×1 matrix read as a scalar (§7.1). For $\mathbf{u} \in \mathbb{R}^m$ and $\mathbf{v} \in \mathbb{R}^n$ the **outer product** $\mathbf{u}\mathbf{v}^\mathsf{T}$ is the $m \times n$ matrix whose (i, j) entry is $u_i v_j$ (§7.2). And $\mathbf{v}^\mathsf{T}\mathbf{v} = \|\mathbf{v}\|^2$ (§7.3).

Problem C.1 (Computing inner and outer products)**.**
Let $\mathbf{u} = [1, -2, 3]$, $\mathbf{v} = [4, 0, -1]$ and $\mathbf{w} = [2, 5]$. Before computing each product, state its shape.

(a) $\mathbf{u}^\mathsf{T}\mathbf{v}$, $\mathbf{v}^\mathsf{T}\mathbf{u}$, $\mathbf{u}^\mathsf{T}\mathbf{u}$ and $\mathbf{v}^\mathsf{T}\mathbf{v}$. Use the last two to find $\|\mathbf{u}\|$ and $\|\mathbf{v}\|$.

(b) $\mathbf{u}\mathbf{v}^\mathsf{T}$ and $\mathbf{v}\mathbf{u}^\mathsf{T}$. Check that $(\mathbf{u}\mathbf{v}^\mathsf{T})^\mathsf{T} = \mathbf{v}\mathbf{u}^\mathsf{T}$.

(c) $\mathbf{u}\mathbf{w}^\mathsf{T}$ and $\mathbf{w}\mathbf{u}^\mathsf{T}$. Is $\mathbf{u}^\mathsf{T}\mathbf{w}$ defined?

(d) Use $\|\mathbf{x}\| = \sqrt{\mathbf{x}^\mathsf{T}\mathbf{x}}$ to compute the norms of $[2, -1, 2, 4] \in \mathbb{R}^4$ and of the all-ones vector $\mathbf{1} \in \mathbb{R}^9$.

Problem C.2 (What an outer product looks like)**.**

(a) Let $\mathbf{u} \in \mathbb{R}^m$ and $\mathbf{v} \in \mathbb{R}^n$. Prove that row i of $\mathbf{u}\mathbf{v}^\mathsf{T}$ is $u_i\mathbf{v}^\mathsf{T}$, and that column j of $\mathbf{u}\mathbf{v}^\mathsf{T}$ is $v_j\mathbf{u}$. So every row is a multiple of the same row vector, and every column is a multiple of the same column vector.

(b) Each matrix below is an outer product $\mathbf{u}\mathbf{v}^\mathsf{T}$. Find a pair $\mathbf{u}$, $\mathbf{v}$ for each, and check it by multiplying out.

$$\begin{bmatrix} 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}, \qquad \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 3 & -3 \end{bmatrix}, \qquad \begin{bmatrix} 0 & 0 & 0 \\ 5 & 10 & -5 \\ -1 & -2 & 1 \end{bmatrix}, \qquad \begin{bmatrix} 6 & -4 & 2 & 8 \end{bmatrix}.$$

Is the pair unique? If not, describe every other pair that works for the first matrix.

(c) Prove that $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ is *not* an outer product $\mathbf{u}\mathbf{v}^\mathsf{T}$ of any two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$.

(d) Prove that $\operatorname{tr}(\mathbf{u}\mathbf{v}^\mathsf{T}) = \mathbf{v}^\mathsf{T}\mathbf{u}$ for all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, where the **trace** of a square matrix is the sum of its diagonal entries (§9.1). Check it on $\mathbf{u}$ and $\mathbf{v}$ from Problem C.1.

(e) Write $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ as the sum of two outer products.

Hint. Try $\mathbf{e}_1$ and $\mathbf{e}_2$ as the column vectors.

Part D. Matrix multiplication

Matrix Unit, §2.3, §2.5 and §8. For $\mathbf{A} \in \mathbb{R}^{m\times p}$ and $\mathbf{B} \in \mathbb{R}^{p\times n}$, the product $\mathbf{AB} \in \mathbb{R}^{m\times n}$ has (i, j) entry

$$(\mathbf{AB})_{ij} = \sum_{k=1}^{p} a_{ik}b_{kj},$$

the dot product of row i of $\mathbf{A}$ with column j of $\mathbf{B}$ (§8.1). It is defined exactly when the number of columns of $\mathbf{A}$ equals the number of rows of $\mathbf{B}$ (§8.2). The laws are **M1**–**M6** of §8.4 ("The algebraic laws"): associativity, the two distributive laws, scalars moving freely, $\mathbf{IA} = \mathbf{AI} = \mathbf{A}$, and $(\mathbf{AB})^\mathsf{T} = \mathbf{B}^\mathsf{T}\mathbf{A}^\mathsf{T}$. There is no commutative law. A matrix–vector product $\mathbf{Ax}$ is the case $n = 1$ (§8.5), and it can be read two ways (§2.5): by rows, its i-th entry is the dot product of row i of $\mathbf{A}$

with $\mathbf{x}$; by columns, it is the linear combination $x_1\mathbf{a}_1 + \cdots + x_p\mathbf{a}_p$ of the columns $\mathbf{a}_1, \ldots, \mathbf{a}_p$ of $\mathbf{A}$. (§2.5 states this for a square matrix; the proof given there works unchanged for any $\mathbf{A} \in \mathbb{R}^{m\times p}$.) Write $\mathbf{e}_i$ for the standard basis vector with a 1 in slot i and 0 elsewhere.

Problem D.1 (Shapes first)**.**
Let $\mathbf{A} \in \mathbb{R}^{2\times 3}$, $\mathbf{B} \in \mathbb{R}^{3\times 3}$, $\mathbf{C} \in \mathbb{R}^{3\times 1}$, $\mathbf{D} \in \mathbb{R}^{1\times 2}$ and $\mathbf{E} \in \mathbb{R}^{2\times 2}$. For each product, say whether it is defined, and if it is, give its shape. Set your answers out as a table.

$$\mathbf{AB},\ \mathbf{BA},\ \mathbf{AC},\ \mathbf{CA},\ \mathbf{BC},\ \mathbf{CB},\ \mathbf{DA},\ \mathbf{AD},\ \mathbf{DE},\ \mathbf{ED},\ \mathbf{EA},\ \mathbf{AE},$$

$$\mathbf{C}^\mathsf{T}\mathbf{B},\ \mathbf{C}\mathbf{C}^\mathsf{T},\ \mathbf{C}^\mathsf{T}\mathbf{C},\ \mathbf{A}^\mathsf{T}\mathbf{E},\ \mathbf{A}\mathbf{A}^\mathsf{T},\ \mathbf{A}^\mathsf{T}\mathbf{A},\ \mathbf{DAC},\ \mathbf{EABC}.$$

Problem D.2 (Multiplying matrices)**.**

(a) Let $\mathbf{E} = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}$ and $\mathbf{F} = \begin{bmatrix} 1 & 5 \\ -2 & 0 \end{bmatrix}$. Compute $\mathbf{EF}$ and $\mathbf{FE}$. Are they equal?

(b) Let $\mathbf{A} = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 3 & 1 \\ 2 & 1 \\ 1 & 0 \end{bmatrix}$. Compute $\mathbf{AB}$ and $\mathbf{BA}$.

(c) Let $\mathbf{G} = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 3 & 0 & 2 \end{bmatrix}$ and $\mathbf{K} = \begin{bmatrix} 2 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 3 & 1 \end{bmatrix}$. Compute $\mathbf{GK}$ and $\mathbf{KG}$.

(d) With $\mathbf{A}$ and $\mathbf{B}$ from (b), compute $\mathbf{B}^\mathsf{T}\mathbf{A}^\mathsf{T}$ and confirm **M6**. Then compute $\mathbf{A}^\mathsf{T}\mathbf{B}^\mathsf{T}$, and say which matrix from (b) it is the transpose of.

(e) Compute $\begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$ and $\begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$, and name each as an inner or an outer product.

Problem D.3 (Matrix–vector products, two ways)**.**

(a) Compute each product twice, by rows and by columns, and check that the two answers agree:

$$\begin{bmatrix} 2 & -1 & 0 \\ 1 & 3 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix}, \qquad \begin{bmatrix} 1 & 4 \\ -2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \qquad \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}.$$

The third product is the zero vector. Write out what the column reading then says about the three columns of that matrix. Are the columns linearly independent? (Use the definition in the table of the vector unit's §24.2.)

(b) Let $\mathbf{A} = \begin{bmatrix} 1 & 0 & 2 \\ 3 & -1 & 1 \\ 0 & 4 & 5 \end{bmatrix}$, with columns $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$. Use the column reading, and no equation solving, to find a vector $\mathbf{x}$ with each property: $\mathbf{Ax} = \mathbf{a}_2$; $\mathbf{Ax} = 3\mathbf{a}_1 - \mathbf{a}_3$; $\mathbf{Ax} = \mathbf{a}_1 + \mathbf{a}_2 + \mathbf{a}_3$. Check the second one by multiplying by rows.

(c) Let $\mathbf{A} \in \mathbb{R}^{m\times n}$. Prove that $\mathbf{A}\mathbf{e}_j$ is column j of $\mathbf{A}$, and that $\mathbf{e}_i^{\mathsf{T}}\mathbf{A}\mathbf{e}_j = a_{ij}$. (Here $\mathbf{e}_j \in \mathbb{R}^n$ and $\mathbf{e}_i \in \mathbb{R}^m$.) Use the second fact to read off $\mathbf{e}_2^{\mathsf{T}}\mathbf{A}\mathbf{e}_3$ for the $\mathbf{A}$ of (b).

(d) Find the 2×2 matrix $\mathbf{M}$ with $\mathbf{M}\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}3\\-1\end{bmatrix}$ and $\mathbf{M}\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}2\\5\end{bmatrix}$. Then compute $\mathbf{M}\begin{bmatrix}4\\-2\end{bmatrix}$ by the column reading.

Problem D.4 (Matrix arithmetic is not number arithmetic)**.**

(a) Let $\mathbf{A} = \begin{bmatrix}1 & 2\\2 & 4\end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix}2 & -4\\-1 & 2\end{bmatrix}$. Compute $\mathbf{AB}$ and $\mathbf{BA}$. For real numbers, $ab = 0$ forces $a = 0$ or $b = 0$. What does (a) show about matrices?

(b) Let $\mathbf{N} = \begin{bmatrix}0 & 1\\0 & 0\end{bmatrix}$ and $\mathbf{N}_3 = \begin{bmatrix}0 & 1 & 0\\0 & 0 & 1\\0 & 0 & 0\end{bmatrix}$. Compute $\mathbf{N}^2$, then $\mathbf{N}_3^2$ and $\mathbf{N}_3^3$. (Here $\mathbf{A}^2 = \mathbf{AA}$ and $\mathbf{A}^3 = \mathbf{AAA}$.)

(c) Use (a) to give matrices with $\mathbf{AB} = \mathbf{AC}$, $\mathbf{A} \neq \mathbf{0}$ and $\mathbf{B} \neq \mathbf{C}$. So a nonzero matrix cannot in general be cancelled.

(d) Let $\mathbf{A}$ and $\mathbf{B}$ be $n\times n$. Prove that

$$(\mathbf{A}+\mathbf{B})^2 = \mathbf{A}^2 + \mathbf{AB} + \mathbf{BA} + \mathbf{B}^2,$$

naming the law of §8.4 used at each step, and deduce that $(\mathbf{A}+\mathbf{B})^2 = \mathbf{A}^2 + 2\mathbf{AB} + \mathbf{B}^2$ holds if and only if $\mathbf{AB} = \mathbf{BA}$. Then compute both sides of the second formula for $\mathbf{E}$ and $\mathbf{F}$ from Problem D.2(a).

Problem D.5 (Products of patterned matrices)**.**
For numbers $d_1, d_2, \ldots, d_n$, the notation $\operatorname{diag}(d_1, d_2, \ldots, d_n)$ means the $n\times n$ diagonal matrix (§4.2) whose diagonal entries are $d_1, d_2, \ldots, d_n$, in that order, and whose other entries are all 0:

$$\operatorname{diag}(d_1, d_2, \ldots, d_n) = \begin{bmatrix} d_1 & 0 & \cdots & 0\\ 0 & d_2 & \cdots & 0\\ \vdots & \vdots & \ddots & \vdots\\ 0 & 0 & \cdots & d_n \end{bmatrix}.$$

(a) Compute $\operatorname{diag}(2,-1,3)\ \operatorname{diag}(4,5,-2)$ and $\operatorname{diag}(4,5,-2)\ \operatorname{diag}(2,-1,3)$.

(b) Let $\mathbf{D} = \operatorname{diag}(2,-1,3)$ and $\mathbf{A} = \begin{bmatrix}1 & 2 & 3\\4 & 5 & 6\\7 & 8 & 9\end{bmatrix}$. Compute $\mathbf{DA}$ and $\mathbf{AD}$. In each product, what happened to the rows or columns of $\mathbf{A}$?

(c) Let $\mathbf{U} = \begin{bmatrix}1 & 2 & 3\\0 & 4 & 5\\0 & 0 & 6\end{bmatrix}$ and $\mathbf{V} = \begin{bmatrix}2 & 0 & 1\\0 & 1 & -1\\0 & 0 & 3\end{bmatrix}$. Compute $\mathbf{UV}$, and observe that it is upper triangular and that its diagonal entries are the products $u_{ii}v_{ii}$. Then prove both facts for any two upper triangular $n\times n$ matrices.

Hint. For $i > j$, look at each term $u_{ik}v_{kj}$ of $(\mathbf{UV})_{ij} = \sum_k u_{ik}v_{kj}$. If $k < i$, which factor is zero? If $k \geq i$, which factor is zero?

(d) Let $\mathbf{S} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Compute $\mathbf{S}^2$, $\mathbf{S}^3$ and $\mathbf{S}^4$, guess a formula for $\mathbf{S}^n$, and check that your formula satisfies $\mathbf{S}\,\mathbf{S}^n = \mathbf{S}^{n+1}$.

(e) Give a formula for $\mathrm{diag}(2, 3)^n$.

Questions about any problem here belong in the class Discord, where everyone can see the answer. Questions that are specific to your own work — "is my proof of D.5(c) valid?" — are better brought to office hours, Thursdays 3:00–4:00 PM in Kiely Hall 508. Please email ahead.